High-level step-by-step system flow

1. Start: User signs in
    - A user opens the system and enters their username and password.
    - The system checks the credentials and figures out the user's role (e.g., Parish Clerk, Parish Priest, Finance HQ, Auditor).
    - If the credentials are valid, the user is taken to the dashboard appropriate for their role. If not, the system shows an error and suggests a password reset.

    Output: User session authenticated, role identified.

2. Budget creation and publication (planning stage)
    - A parish or department prepares a budget draft for a time period (month / quarter / year). Draft includes budget lines (categories) and planned amounts (e.g., Maintenance: 1,000, Construction: 5,000).
    - The clerk saves the draft in the system. The system records who created it and when.
    - The parish leadership reviews and submits the draft to Archdiocese Finance HQ for approval. Submitting changes status from Draft to Submitted.
    - Finance HQ reviews the submitted budget:
        - If acceptable, Finance HQ marks it Approved and the system publishes the approved budget for that parish and period.
        - If not acceptable, Finance HQ rejects it with comments; parish revises the draft and re-submits.

    Outputs: Approved budgets published; draft/resubmission loop logged.

3. Budget availability & monitoring
    - Once a budget is approved, the system creates budget lines with available amounts equal to planned amounts.
    - The dashboard shows budget totals and remaining amounts by category, including simple indicators (e.g., green/amber/red) for utilization.
    - The system constantly updates "available" as income and expenses are recorded.

    Output: Live budget availability shown to users.

4. Recording income (money in)

- A finance clerk records a new income transaction: amount, date, source (mass collection, donation, grant, rent), parish, and optional category or budget line.

- The clerk may upload supporting evidence (receipt, bank confirmation).

- The system stores the record, attaches the document, and logs who recorded it.

- If the income is designated for a budget line, the system adds the amount to that budget line's available balance. If unrestricted, it records as general income.

Outputs: Income transaction saved; budget updated (if applicable); audit entry created.

5. Recording expenses (money out)
    - A finance clerk creates an expense entry: amount, date, payee, parish/department, expense category, description, and attaches invoices or receipts.

    - The system performs an automatic check against the relevant budget line(s):

        - If the expense ≤ available amount → the system allows posting or automatically approves (depending on rules), deducts amount from budget available, marks the transaction as Approved or Posted, and logs the action.

        - If the expense > available amount → the system creates an Approval Case and marks the expense Pending Approval. It notifies the defined approvers (parish leadership and/or Finance HQ).

        - Approver(s) review the pending expense and:

            - Approve: system marks expense Approved, deducts the budget, and logs the approval.

            - Reject: system marks Rejected with comments; the clerk can correct and re-submit or cancel.

    Outputs: Expense saved; budget adjusted or approval workflow started; audit entry.

6. Approval escalation rules (exceptions)
    - The system uses configurable thresholds (e.g., amounts above X must be approved by Parish Priest; amounts above Y by Finance HQ).

    - When approval is needed, the system creates a task and sends notifications (email/app) to approvers with transaction details and attachments.

- The approver takes action in the system: Approve, Reject, Request more info. All decisions are recorded.

Output: Transparent approval history and decisions logged.

7. Document management and attachments
    - All documents (titles, receipts, invoices, contracts, land documents) are uploaded and linked to the relevant records (asset, expense, income).
    - Each upload stores metadata: who uploaded it, when, associated record, and a checksum/hash for integrity.
    - The system provides a simple viewer and ensures only permitted users can download or edit documents.

    Output: Secure, auditable document repository tied to transactions.

8. Asset management (parallel flow)
    - Users add or edit assets: type (land, building), location, value, ownership details and upload title documents.
    - The system stores asset records and keeps a change history (who edited what and when).
    - Asset valuation updates can be recorded and time-stamped.
    - Asset records can be used by reporting and for audits.

    Output: Complete asset registry with audit trail.

9. Budget vs Actual monitoring and alerts
    - The system calculates live Budget vs Actual figures per parish, department, and consolidated across the Archdiocese.
    - Thresholds trigger alerts: e.g., when a budget line reaches 70% (warning), 90% (critical), or goes negative (overspent).
    - Alerts are shown on dashboards and sent to nominated users.

    Output: Proactive notifications to avoid overspend; trending indicators.

10. Reporting and exports

- Authorized users select report parameters (parish, date range, category, level of detail).

- The system compiles data (income, expenses, budgets, variances) and generates previews.

- Users can export reports as PDF, Excel, or CSV. The system stores report snapshots for future reference.

- Auditors can export audit trail and transaction attachments for verification.

Output: Exportable, time-stamped reports and snapshots.

11. Audit logs and compliance
    - Every important action (create, edit, approve, delete) is logged with user, timestamp, record affected, and explanation.

    - Logs are immutable (cannot be altered by normal users) and are available to auditors.

    - Regular backups of DB and documents are stored as per the Archdiocese policy.

    Output: Immutable audit trail and backup records.

12. Month-end / Year-end close
    - At each month or year end, users run final reports and reconcile income/expense with bank statements and other records.

    - The system can lock closed periods (no more edits) or require exceptional approval for post-close adjustments.

    - Finalized reports are archived.

    Output: Closed financial periods and archival for auditing.

13. Error handling & exceptions
    - If a transaction fails validation (missing fields, bad file format), the system shows a clear error and prevents save.

    - For system failures (server/db down), the system shows friendly maintenance messages and queues incoming activity where possible.

    - Any suspicious activity triggers an immediate alert to administrators.

Output: Controlled error messages, queued work, and security alerts.

14. Notifications & communication
    - System notifies users by email and/or in-app messages for key events:
        - budget approval/rejection,
        - expense needing approval,
        - near-budget alerts,
        - system maintenance notices.
        - Notification preferences are configurable per user/role.

    Output: Timely notifications and audit of who was notified.

15. End: Reporting to stakeholders & follow-up
    - Stakeholders review dashboards and exported reports.
    - If issues are found, they open follow-up actions: investigations, adjustments, or future budget changes.
    - All follow-ups are tracked in the system as tasks or notes.

    Output: Continuous improvement loop for budgeting and controls.

1. Quick examples of common scenarios (so flow is concrete)
    a. Scenario A — Small expense within budget
        i. Clerk records expense of 50 that fits available budget.
        ii. System auto-approves, deducts 50 from budget line, logs action.
        iii. Dashboard reflects new available amount.
    b. Scenario B — Large expense above budget
        i. Clerk records expense of 10,000 that exceeds budget by 6,000.
        ii. System marks expense Pending Approval and notifies approver(s).
        iii. Approver reviews attachments, approves → system deducts amount and logs approval. — or — Approver rejects → clerk revises or cancels.
    c. Scenario C — Income designated to project
        i. Donation of 5,000 is recorded and tagged to “School Roof” budget line.
        ii. System increments that budget line’s available amount by 5,000.
        iii. Project now has additional funds available for spending.
        iv. Short checklist for developer acceptance (what each step must produce)
        v. Authentication produces token + role.

vi. Budget creation produces Draft + budget lines with planned & available amounts.
vii. Budget approval flips status and publishes lines as active.
viii. Income/Expense create ledger entries, attachments stored, and audit log entry created.
ix. Budget validation runs on expense create; approval workflow created if needed.
x. Reports aggregate budget vs actual and allow export.
xi. Audit log is immutable and searchable.
xii. Backup job runs daily/weekly as required.